

L4.1b Graphing Functions of y

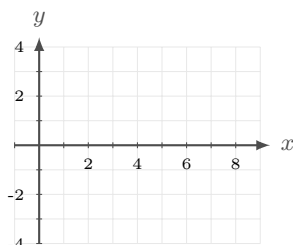
Work in your group. Check your answers against the Quarto tonight — the graphing trainer in it has every curve here except item 7(b).

Graphing Sideways

The whole method. Pick a value of y . Compute x . Plot the point (x, y) — **the table reads y first, but the point is still written x first**. Items 1–2 build the curve from the table. Items 3–7 use what you already know about the family; the table is only your check.

- Sketch $x = 2^y$. Plotting $y = \log_2 x$ directly means answering “2 to what power gives 5?” at every point. Written as $x = 2^y$, you choose y instead.

y	x
-2	
-1	
0	
1	
2	
3	

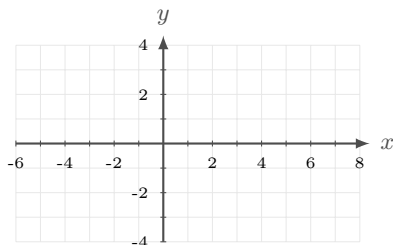


(a) $y = 2^x$ has a horizontal asymptote as $x \rightarrow -\infty$ and a y -intercept at $(0, 1)$. Name the matching feature of *this* curve, and say where it sits.

(b) Same curve both times, and nothing was reflected. In one line: what did change?

- Sketch $x = 2y + 1$.

y	x
-2	
-1	
0	
1	
2	
3	



(a) The coefficient of y is 2. It is still rise over run — but which is which now? Fill in: $2 = \frac{\Delta \text{ }}{\Delta \text{ }}$ so from any point on the line, go **up** and **across** .

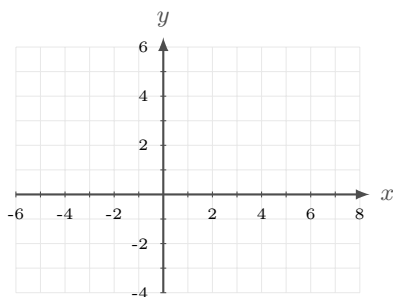
(b) The constant 1 is an **x -intercept**, at $(1, 0)$ — $g(0)$ lands on the **horizontal** axis now. Mark it, then step off the slope to get your second point.

(c) Solve for y . The coefficient now reads $\frac{1}{2}$. **Reconcile the two in one line.**

(d) Is this curve monotonic? How do you know?

3. Sketch $x = y^2 - 2y - 3$.

y	x
-2	
-1	
0	
1	
2	
3	
4	



(a) Set $x = 0$ and factor **by Loh**: the roots average $m = \underline{\hspace{1cm}}$, so they are $m \pm u$; their product is -3 , so $u = \underline{\hspace{1cm}}$, and the roots are $\underline{\hspace{2cm}}$.

(b) Those roots are the **y**-intercepts. Mark them on the **vertical** axis.

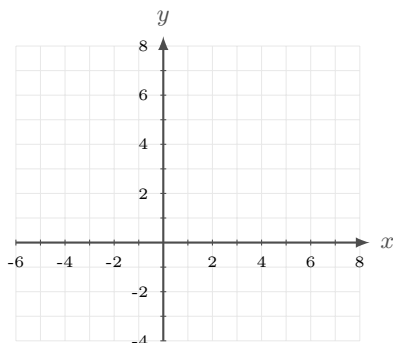
(c) m is the **horizontal** line $y = \underline{\hspace{1cm}}$ — the axis of symmetry. Where is the vertex?

(d) Which way does it open, and how do you know from the equation alone?

(e) Now complete the square in y and solve for y : you should get $y = 1 \pm \sqrt{x+4}$, two square-root branches. **Which branch is which half of your sketch?** Then say in one line why the y -table never made you choose between them.

4. Sketch $x = \frac{y}{y-2}$.

y	x
-2	
0	
1	
1.5	
2.5	
3	
4	
6	



(a) The denominator is 0 at $y = 2$. As $y \rightarrow 2$, what happens to x ? So _____ is an asymptote — **vertical** or **horizontal**?

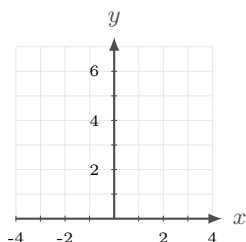
(b) Compute $\lim_{y \rightarrow \pm\infty} \frac{y}{y-2}$. So _____ is an asymptote — **vertical** or **horizontal**?

(c) Both came out opposite to the rule you know for $y = f(x)$. Say in one line what actually flipped — and what did **not** move at all.

(d) Mark the zero as well. Those three features — the two asymptotes and the zero — pin the function completely. Write $x = \frac{a(y-r)}{y-p}$ and fill in a , r , p from your graph.

5. Sketch $x = 2 \sin y$ for $0 \leq y \leq 2\pi$.

y	x
0	
$\pi/6$	
$\pi/2$	
$5\pi/6$	
π	
$3\pi/2$	
2π	



(a) Zeros first. $\sin y = 0$ whenever $y = n\pi$, so mark 0, π , 2π on the **vertical** axis. Then the curve between them is easy to draw.

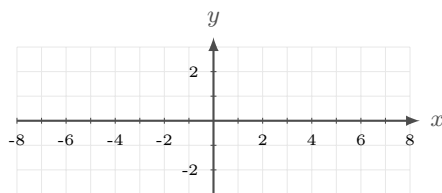
(b) The turning points sit halfway between the zeros. Confirm with $\frac{dx}{dy} = 0$: $y =$ _____. You are differentiating x with respect to y — same rules, other letter.

(c) Split $[0, 2\pi]$ into intervals on which the curve is monotonic. How many?

(d) A **horizontal** line at $y = \pi/4$ meets the curve _____ time(s). A **vertical** line at $x = 1$ meets it _____ time(s). Which count is the one that lets you write x as a function of y ?

6. Two cubics — sketch both on the one grid. (i) $x = y^3$ (ii) $x = y^3 - 3y$

y	(i) y^3	(ii) $y^3 - 3y$
-2		
-1		
0		
1		
2		



(a) One is monotonic and one is not. Which — and what in the table shows it?

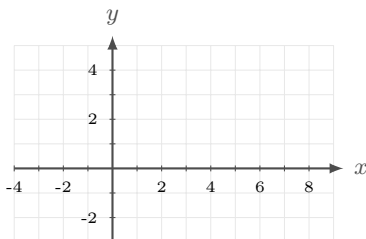
(b) For the one that is not, find its turning points from $\frac{dx}{dy} = 0$. You are differentiating x with respect to y — same rules, other letter.

(c) Write the three intervals of y on which (ii) **is** monotonic.

(d) Curve (i) is exactly how you would sketch $y = \sqrt[3]{x}$ without ever computing a cube root. Explain in one sentence.

7. Transformations — **the rule you already have, unchanged.** *Outside the function moves the output. Inside moves the input, backward.* Here the output is x (across) and the input is y (up).

(a) $x = 3(y - 1)^2 - 1$, from the parent $x = y^2$.



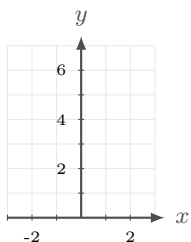
$(y - 1)$ is **inside**, so it moves the **input** backward: the whole curve shifts _____ by ____.

The 3 and the -1 are **outside**, so they move the **output** normally: a stretch _____ and a shift _____.

Vertex at (____, ____). Axis of symmetry: the **horizontal** line $y = \underline{\hspace{1cm}}$. Opens ____.

Say what is different from $y = 3(x - 1)^2 - 1$ — and what is the same.

(b) $x = \cos\left(2y - \frac{\pi}{2}\right)$, from the parent $x = \cos y$.



1 · Anchor it. Set the inside to zero: $2y - \frac{\pi}{2} = 0$ gives $y = \underline{\hspace{1cm}}$ — where the parent cosine starts, at its maximum.

2 · Period $= 2\pi/2 = \underline{\hspace{1cm}}$, so the turning points come every **half-period**: $y = \underline{\hspace{1cm}}$

3 · Zeros sit a quarter-period off each turning point: $y = \underline{\hspace{1cm}}$

Amplitude _____, measured **across**, not up.

$\cos y$ is even. Is this one? Use $\cos\left(\theta - \frac{\pi}{2}\right) = \sin \theta$ to rewrite it, then test $x(-y)$.

Monotonic pieces

8. $y = x^2 + 2x$ on $[-3, 2]$.

Stop. What does a horizontal line have to do on a piece of the graph before you can write x as a function of y there?

- (a) Sketch it, and draw one horizontal line that crosses it twice.
- (b) Find the turning point from $f'(x) = 0$.
- (c) Split $[-3, 2]$ into intervals on which x *can* be written as a function of y , and write x in terms of y on each one.